

Database Systems (CS2005)

Sessional-I Exam

Course Instructor(s):

Asif Malik

Section(s): SE-A, SE-B

Total Time (Hrs): 1

Total Marks: 64

Total Questions: 6

Date: Feb 22, 2025

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

Q1: Select the appropriate answer in the following by encircling them.

[25 marks]

1. Which of the following is the benefit of data independence in a DBMS?
 - a) It simplifies the process of data retrieval.
 - b) It allows data to be accessed from any device.
 - c) It reduces data redundancy.
 - d) It insulates application programs from changes in the data structure.**
2. What is the primary function of a database administrator (DBA)?
 - a) Designing user interfaces for database applications
 - b) Writing SQL queries for end-users
 - c) Managing and maintaining the database system**
 - d) Developing web-based applications
3. In a multi-user database system, what is concurrency control?
 - a) A security feature
 - b) The ability to process queries quickly
 - c) The ability to access data from multiple devices
 - d) The management of simultaneous access to the database by multiple users**
4. What does metadata refer to in a database system?
 - a) Data about data, describing the structure and organization of the database**
 - b) Data that is stored in binary format
 - c) Data that is encrypted for security
 - d) Data that is not relevant to the database
5. What is a composite attribute in the ER model?
 - a) An attribute that can be divided into smaller sub-attributes**
 - b) An attribute that is a primary key
 - c) An attribute that is part of a relationship
 - d) An attribute that cannot be null
6. What does the cardinality of a relationship in the ER model indicate?
 - a) The total number of entities in the database
 - b) The maximum number of entity instances that can be involved in the relationship**
 - c) The number of attributes in an entity
 - d) The total number of attributes in the database
7. In ER modeling, what does the term "relationship" represent?

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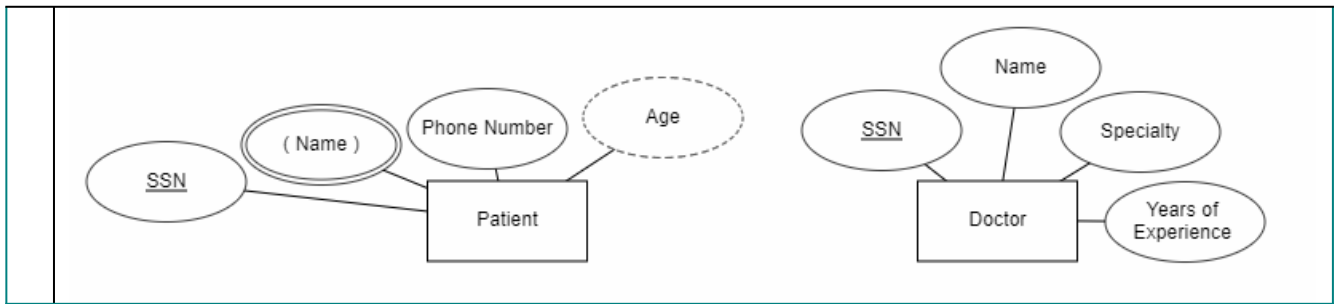
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- a) A foreign key constraint
 - b) A data attribute
 - c) A meaningful link between entities**
 - d) A primary key constraint
8. What is a weak entity in the ER model?
- a) An entity with a strong primary key
 - b) An entity that has a multivalued attribute
 - c) An entity that cannot be uniquely identified by its attributes alone**
 - d) An entity that has no attributes
9. Patients are identified by an SSN, and their names, addresses, and ages must be recorded. Doctors are identified by an SSN. For each doctor, the name, specialty, and years of experience must be recorded. Every patient has a primary physician.

a	
b	
c	
d	
e	

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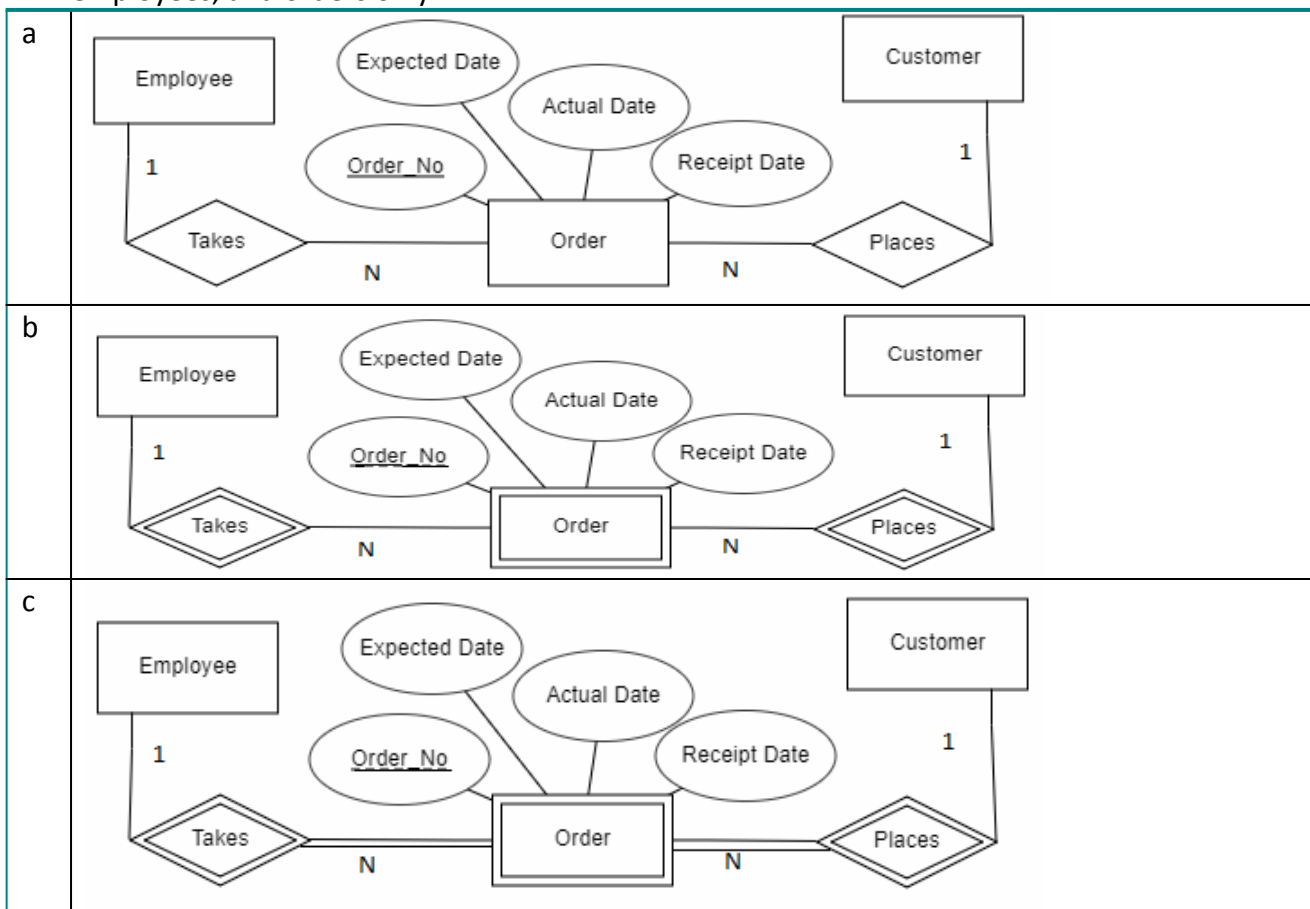


Q10-11

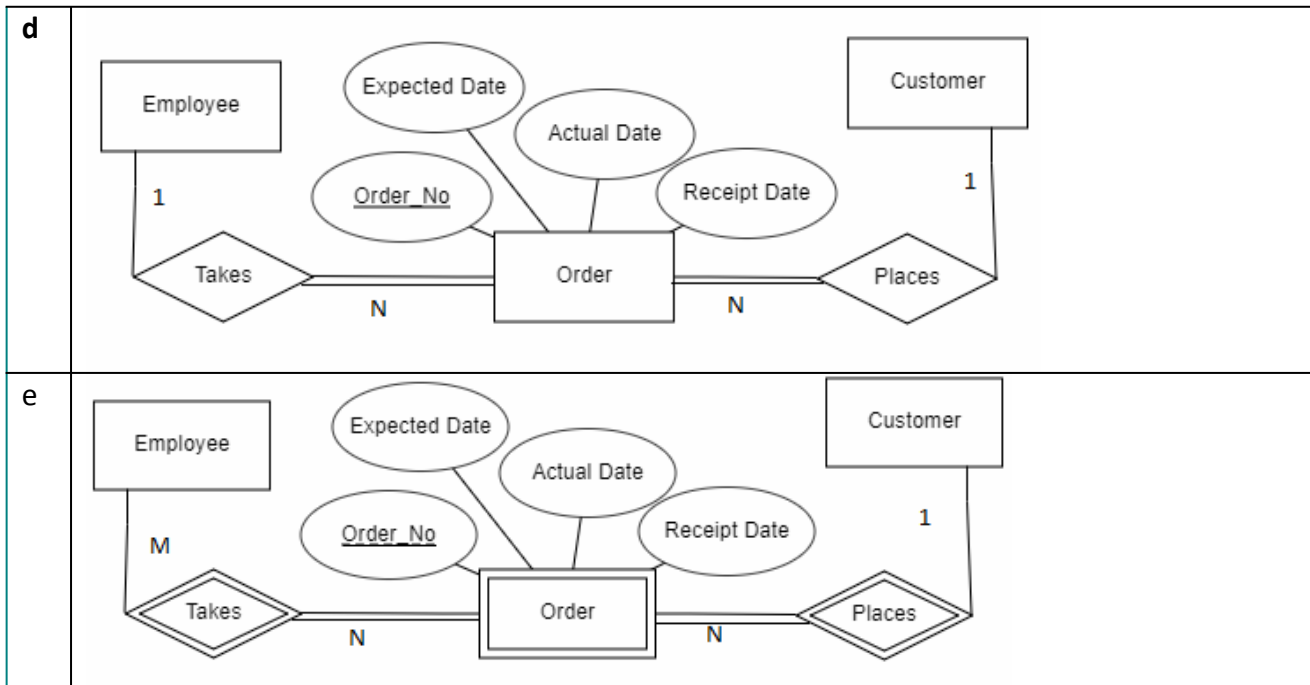
Consider a MAIL_ORDER database in which employees take orders for parts from customers. The data requirements are summarized as follows:

- The mail-order company has employees, each identified by a unique employee number, first and last name, and Zip Code.
- Each customer of the company is identified by a unique customer number, first and last name, and Zip Code.
- Each part sold by the company is identified by a unique part number, a part name, price, and quantity in stock.
- Each order placed by a customer is taken by an employee and is given a unique order number. Each order contains specified quantities of one or more parts. Each order has a date of receipt as well as an expected ship date. The actual ship date is also recorded.

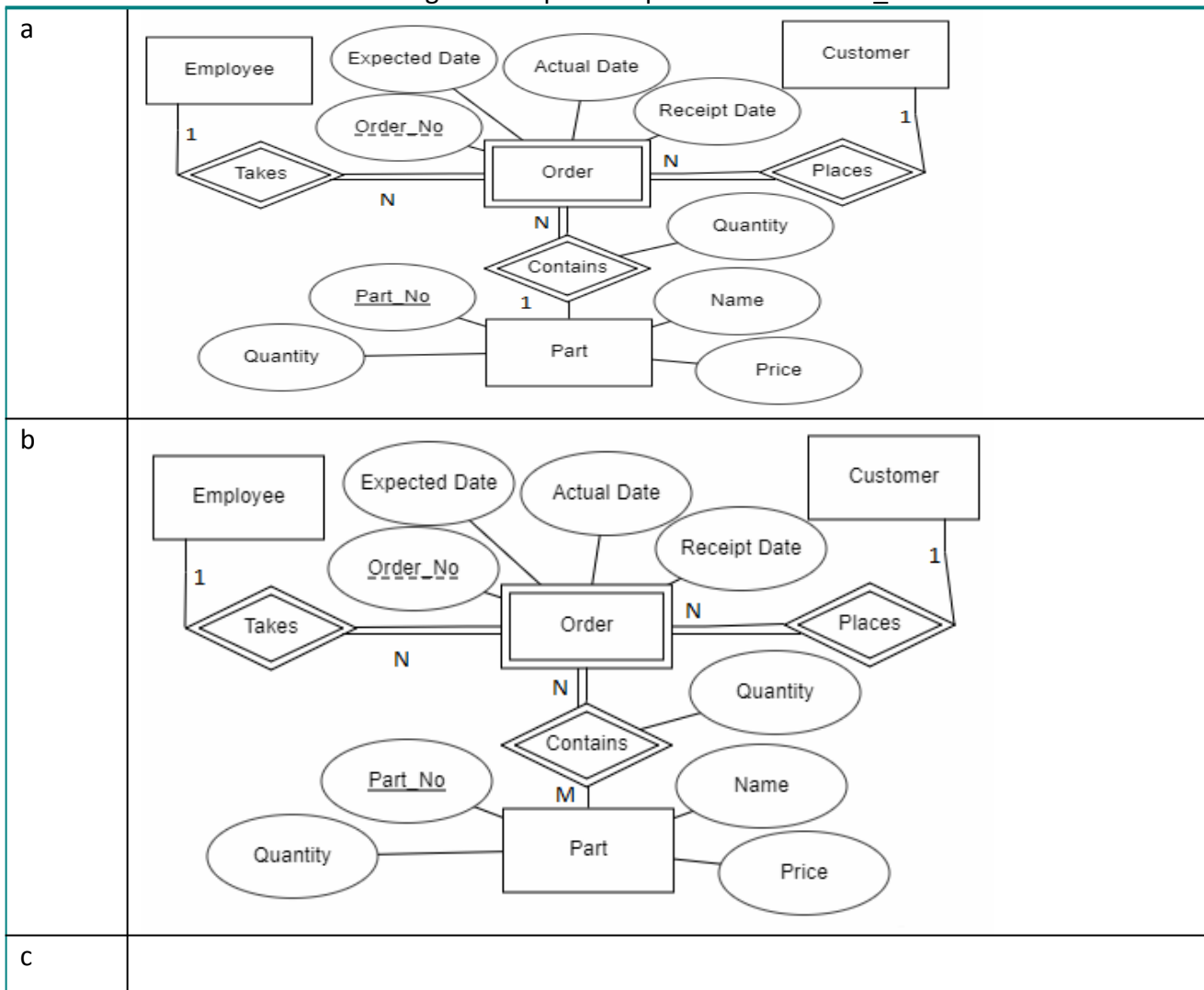
10. Select correct the conceptual schema considering the relationship between customers, employees, and orders only.



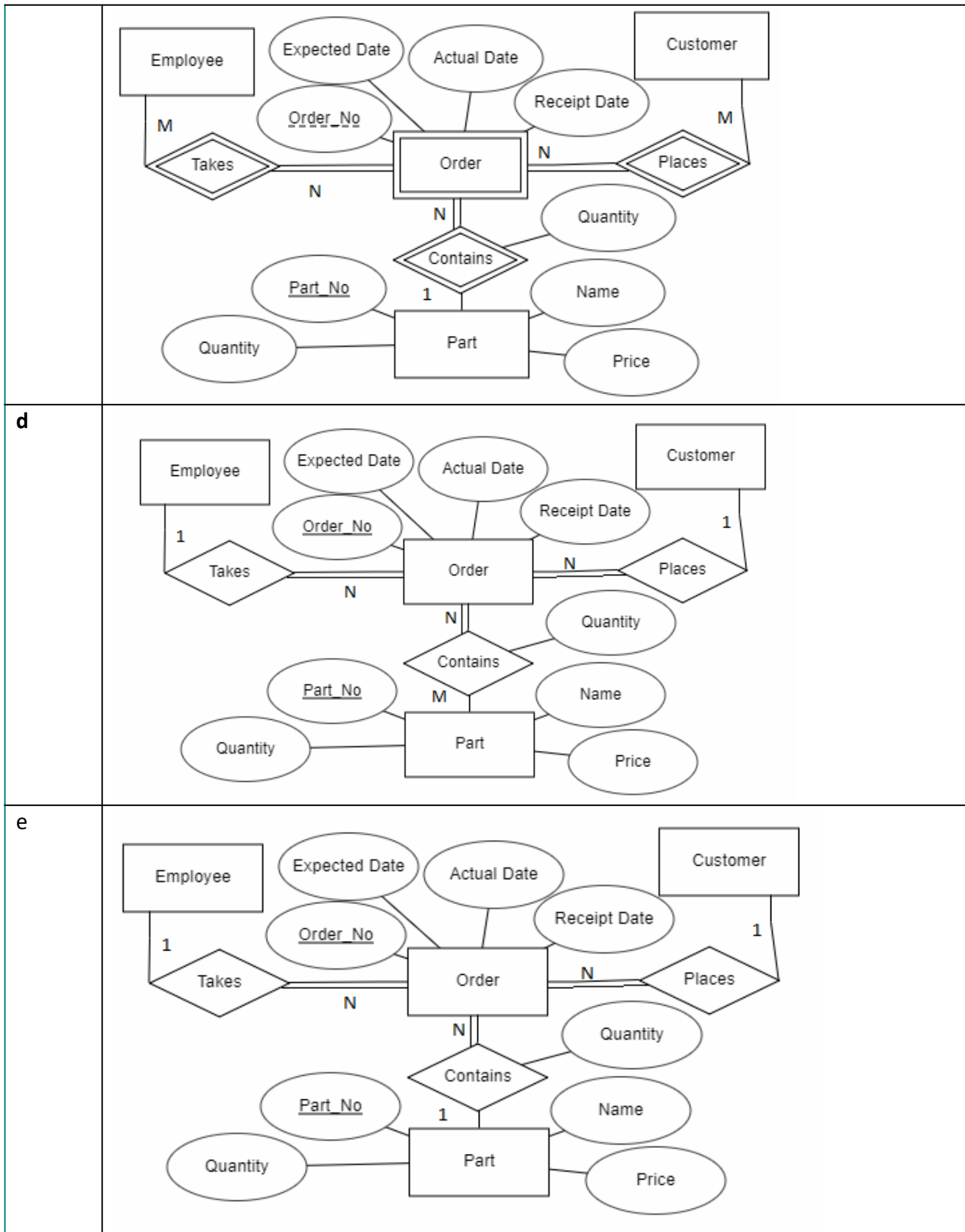
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11. Select the correct ERD covering the complete requirements of MAIL_ORDER database



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12. Each Player may play in a 0 to 1 team, and each Team should have many Players. All teams must have players and a player may not have a team.

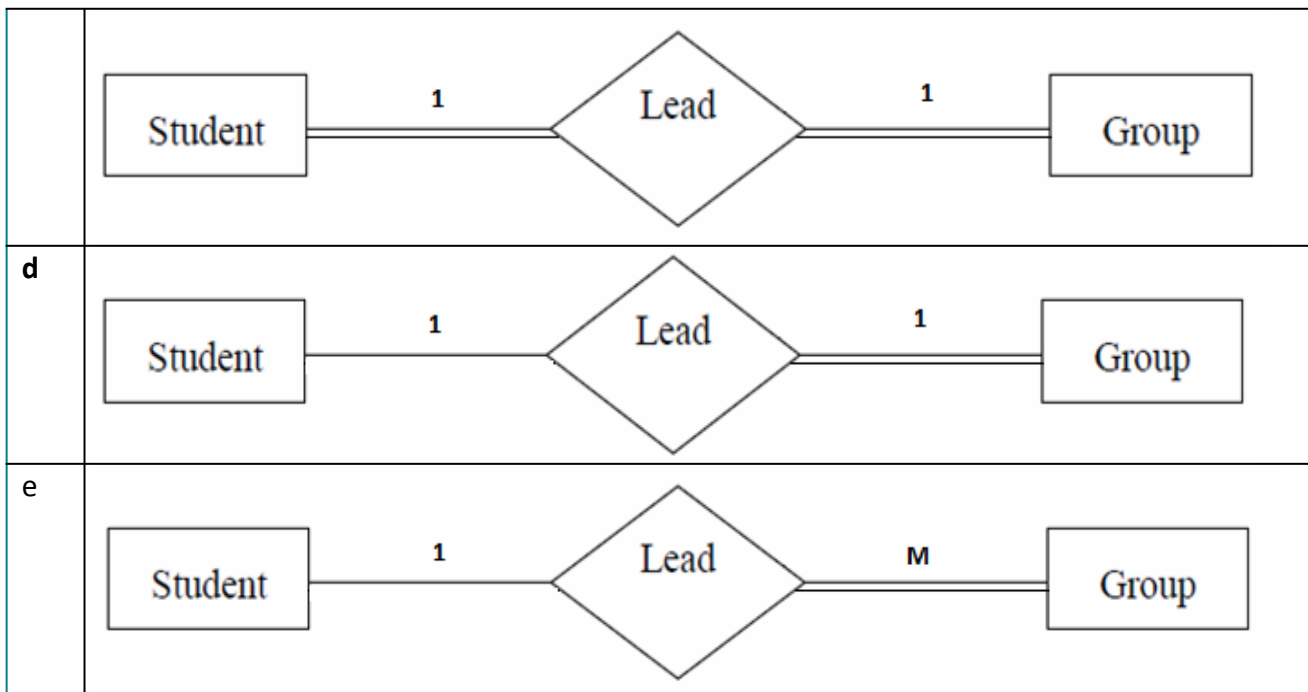
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a	<pre> graph LR Player[Player] --- 1 have{have} have --- M Team[Team] </pre>
b	<pre> graph LR Player[Player] === 1 have{have} have === M Team[Team] </pre>
c	<pre> graph LR Player[Player] === M have{have} have === 1 Team[Team] </pre>
d	<pre> graph LR Player[Player] === M have{have} have -.- 1 Team[Team] </pre>
e	<pre> graph LR Player[Player] === M have{have} have -.- 1 Team[Team] </pre>

13. Each Student may lead 0 to 1 Group, and each Group should be led by only one Student. Not all students can lead groups. All groups must be led.

a	<pre> graph LR Student[Student] --- 1 Lead{Lead} Lead --- 1 Group[Group] </pre>
b	<pre> graph LR Student[Student] === 1 Lead{Lead} Lead === 1 Group[Group] </pre>
c	

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14. Which of the following refers to the level of data abstraction that describes exactly how the data is stored?

- a) Conceptual Level
- b) Logical Level
- c) File Level
- d) Physical Level**
- e) None of the above

For questions (15 – 20) that follow, refer to the figure that shows the ER Diagram of an Airline Database system. Choose the best option for answering the given MCQs.

15. Which of the following is a key attribute in AIRPLAN_TYPE?

- a) Seat_no
- b) Code
- c) Date
- d) Type_name**
- e) All of the above

16. Which of the following is not true about Flight Legs:

- a) A single flight leg must depart from one airport only
- b) A single flight leg must arrive to one airport only
- c) Many flights may depart from the same airport
- d) A flight may not depart from any airport**
- e) None of the above

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17. All of the following are weak entities except:
- a) Fare
 - b) Seat
 - c) Leg Instance
 - d) Flight Leg
 - e) None of the Above**
18. Which of the following is a multi-valued attribute:
- a) Seat_no
 - b) Airplane_id
 - c) Cphone**
 - d) Weekdays
 - e) None of the above
19. Which of the following is an example of a Recursive Relationship
- a) Can_Land
 - b) Assigned
 - c) Departs,Arrives
 - d) Legs,Fares
 - e) None of the Above**
20. What is not true about LEG_INSTANCE -----DEPARTS----- AIRPORT
- a) In the DEPARTS binary relationship type, AIRPORT:LEG_INSTANCE is of cardinality ratio 1:N
 - b) Each airport can be related to any number of LEG_INSTANCES
 - c) A LEG_INSTANCE can be related to (departs from) at most one AIRPORT
 - d) For this particular relationship type DEPARTS, a particular AIRPORT entity can be related to any number of LEG_INSTANCES.
 - e) A LEG_INSTANCE can be related to any number of AIRPORTs.**
21. Information is known as recorded facts while data is an understandable specific representation of data.
- a) True
 - b) False**
22. The data in the database at a particular moment in time is called a database _____
- a) Schema
 - b) Instance**
 - c) Both a and b
 - d) None of the above
23. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____ integrity constraint.
- a) Entity
 - b) Referential**
 - c) Attribute
 - d) Domain
24. The ability to query data, as well as insert, delete, and alter tuples, is offered by _____

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- a) TCL (Transaction Control Language)
- b) DML (Data Manipulation Language)**
- c) DCL (Data Control Language)
- d) DDL (Data Definition Language)

25. What are rows of a relation known as?

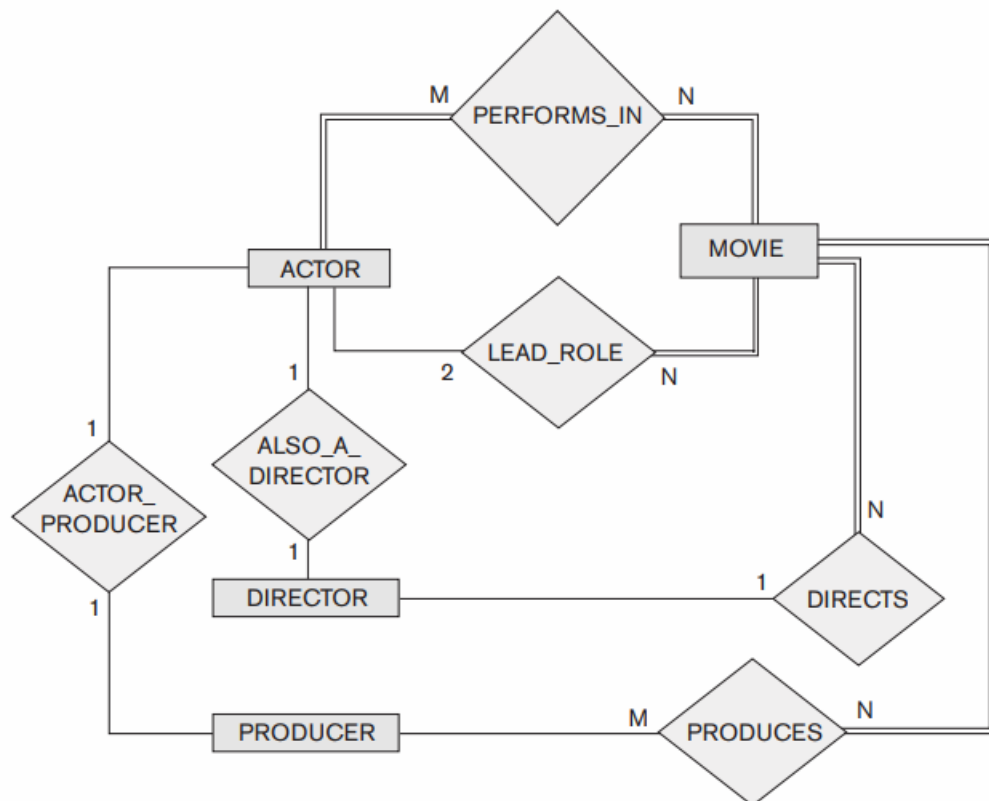
- a) Degree
- b) Tuple**
- c) Entity
- d) None of the above

[CLO 2: Design a conceptual model using (E)ER diagrams for an enterprise]

Q2:

[07 marks]

Assume the Following ER Schema Given the constraints shown in the ER schema, respond to the following statements with **True**, **False**, or **Maybe**. Assign a response of Maybe to statements that, although not explicitly shown to be True, cannot be proven False based on the schema as shown. Justify each answer.



- a) There are no actors in this database that have been in no movies. __TRUE__
- b) There are some actors who have acted in more than ten movies. __TRUE__
- c) Some actors have done a lead role in multiple movies. __TRUE__
- d) A movie can have only a maximum of two lead actors. __TRUE__

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- e) A producer cannot be an actor in some other movie. _____ FALSE _____
- f) There are movies with more than a dozen actors. _____ MAY BE _____
- g) Some producers have been a director as well. _____ MAY BE _____

[CLO 1: Describe the storage and retrieval mechanism in different databases]

Q3: Briefly Explain the following your answer should not exceed three lines. **[2 + 2 + 2 = 06 marks]**

- a) What is the difference between a database schema and a database state?

- b) What is meant by a recursive relationship type? Give some examples of recursive relationship types

- c) What is an entity type? What is an entity set? Explain the differences among an entity, an entity type, and an entity set

[CLO 2: Design a conceptual model using (E)ER diagrams for an enterprise]

Q4: **[10 marks]**

Cardinality ratios often dictate the detailed design of a database. The cardinality ratio depends on the real-world meaning of the entity types involved and is defined by the specific application. For the following binary relationships, suggest cardinality ratios based on the common-sense meaning of the entity types. Clearly state any assumptions you make.

Entity 1	Cardinality Ratio	Entity 2	Justification
COUNTRY	___1:1___	CURRENT_PRESIDENT	

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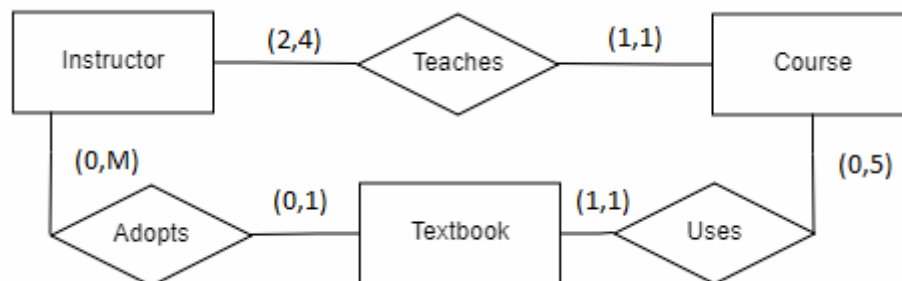
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COURSE ____ M:N ____ TEXTBOOK	
EBAY_AUCTION_ITEM ____ 1:M ____ EBAY_BID	
STUDENT ____ M:1 ____ UNIVERSITY	
BOOK ____ M:N ____ AUTHOR	
CUSTOMER ____ 1:M ____ PURCHASE	
AIRLINE ____ M:N ____ PASSENGER	
DOCTOR ____ M:N ____ PATIENT	
BANK ____ M:N ____ ACCOUNT_HOLDER	
POLICE_STATION ____ 1:M ____ POLICE_OFFICER	

[CLO 2: Design a conceptual model using (E)ER diagrams for an enterprise]

Q5: Answer the following questions with respect to the ER diagram below

[10 marks]



a) How is the relationship between a course and a textbook defined in this scenario?

- Each course can use between 0 and 5 textbooks, and each textbook must be used by exactly one course.

b) What are the limitations on the number of textbooks a course can use?

A course can use a minimum of 0 and a maximum of 5 textbooks.

c) What is the range of courses an instructor can teach?

An instructor can teach between 2 and 4 courses.

d) Can a textbook be used by more than one course?

No, a textbook can only be used by one course, as indicated by the (1,1) cardinality on the textbook side of the "Uses" relationship.

e) How is the adoption of a textbook by an instructor defined?

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An instructor can adopt between 0 and M textbooks, while a textbook can be adopted by at most 1 instructor.

f) Is it necessary for an instructor to adopt a textbook for all the courses they teach?

No, it is not necessary for an instructor to adopt a textbook for all the courses they teach. The "Adopts" relationship is independent of the "Teaches" relationship.

g) What is the minimum and maximum number of courses an instructor can teach?

The minimum number of courses an instructor can teach is 2, and the maximum is 4.

h) How is the adoption of a textbook related to the instructor teaching a course?

There is no direct relationship between the adoption of a textbook by an instructor and the courses they teach. The "Adopts" relationship is between an instructor and a textbook, while the "Uses" relationship connects courses to textbooks.

i) Is there any requirement for a course to have a textbook?

No, a course is not required to have a textbook, as indicated by the (0,5) cardinality in the "Uses" relationship.

j) How does the existence of a textbook relate to its usage in a course?

A textbook must be used by exactly one course (as indicated by the (1,1) cardinality on the textbook side of the "Uses" relationship). However, a course is not required to use a textbook.

[CLO 5: Author queries using relational algebra and SQL]

Q6: Write SQL queries to achieve following:

[06 marks]

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EMPLOYEE

FNAME	MINIT	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
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DEPARTMENT

DNAME	<u>DNUMBER</u>	MGRSSN	MGRSTARTDATE
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DEPT_LOCATIONS

<u>DNUMBER</u>	<u>DLOCATION</u>
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PROJECT

PNAME	<u>PNUMBER</u>	PLOCATION	DNUM
-------	----------------	-----------	------

WORKS_ON

<u>ESSN</u>	<u>PNO</u>	HOURS
-------------	------------	-------

DEPENDENT

<u>ESSN</u>	<u>DEPENDENT_NAME</u>	SEX	BDATE	RELATIONSHIP
-------------	-----------------------	-----	-------	--------------

- a) Retrieve the first name, last name, and salary of employees who earn more than 50,000 and belong to department 3.

```
SELECT first_name, last_name, salary
FROM EMPLOYEE
WHERE salary > 50000 AND department_id = 3;
```

- b) Retrieve the first name, last name, and department name of all employees by joining the EMPLOYEE and DEPARTMENT tables.

```
SELECT e.first_name, e.last_name, d.department_name
FROM EMPLOYEE e
JOIN DEPARTMENT d ON e.department_id = d.department_id;
```

- c) Retrieve the first name, last name, department name, and project name of employees who are working on a project.

```
SELECT E.FNAME, E.LNAME, D.DNAME, P.PNAME
FROM EMPLOYEE E
JOIN DEPARTMENT D ON E.DNO = D.DNUMBER
```

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```
JOIN WORKS_ON W ON E.SSN = W.ESSN  
JOIN PROJECT P ON W.PNO = P.PNUMBER;
```

- d) Retrieve employees along with their supervisor's names, but only for employees whose department number is greater than 5. Order the results by the employee's last name.

```
SELECT E.FNAME AS Employee_FirstName,  
       E.LNAME AS Employee_LastName,  
       S.FNAME AS Supervisor_FirstName,  
       S.LNAME AS Supervisor_LastName  
FROM EMPLOYEE E  
JOIN EMPLOYEE S ON E.SUPERSSN = S.SSN  
WHERE E.DNO > 5  
ORDER BY E.LNAME;
```

- e) Retrieve the first name and last name of all employees who work in department 3 or 5, without using IN operator.

```
SELECT FNAME, LNAME  
FROM EMPLOYEE  
WHERE DNO = 3  
  
UNION  
  
SELECT FNAME, LNAME  
FROM EMPLOYEE  
WHERE DNO = 5;
```

- f) Retrieve the first name, last name, and salary of employees who belong to the department with the accounts department.

```
SELECT E.FNAME, E.LNAME, E.SALARY  
FROM EMPLOYEE E  
JOIN DEPARTMENT D ON E.DNO = D.DNUMBER  
WHERE D.DNAME = 'Accounts';
```